

Predicting Air Quality in Italy Using Fuzzy and Neuro-Fuzzy Modelling

INTELLIGENT SYSTEMS

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1 Introduction

Rapid urban growth, industrial expansion, and increasing energy consumption have significantly altered the composition of the atmosphere. As populations rise and human activities intensify, air pollution has emerged as one of the most pressing environmental challenges of the modern era. Emissions from vehicles, industries, and other anthropogenic sources release a complex mix of pollutants, degrading air quality and threatening public health, ecosystems, and climate stability.

Accurate air quality forecasting plays a crucial role in minimizing these impacts by enabling early warnings and supporting data-driven environmental management. However, modeling air pollution dynamics remains challenging due to the inherent uncertainty and nonlinearity present in environmental and sensor data. Classical regression and time-series models often struggle to manage missing values, noise, and sensor drift, common issues in real-world monitoring systems. To address these limitations, this project explores the application of artificial intelligence (AI) and soft computing techniques capable of learning complex, nonlinear relationships without relying on rigid mathematical assumptions.

Fuzzy logic and neuro-fuzzy systems provide effective tools for air quality prediction. Fuzzy models manage uncertainty through interpretable linguistic rules, making them robust to noise and incomplete data. When combined with the adaptive learning capability of neural networks, neuro-fuzzy systems, such as the Adaptive Neuro-Fuzzy Inference System (ANFIS), create a hybrid framework capable of self-tuning parameters and capturing nonlinear dependencies more effectively than classical statistical models. These characteristics make them particularly suitable for environmental datasets, where variable interactions are complex and time-dependent.

This project applies fuzzy and neuro-fuzzy methods

to predict pollutant concentrations using the Air Quality Dataset from the UCI Machine Learning Repository, which contains 9,358 hourly measurements collected between March 2004 and February 2005. The dataset includes readings from metal-oxide gas sensors, meteorological variables, and reference concentrations of CO, NO_x, NO₂, and benzene. The project involves several key stages: preprocessing and cleaning the data to handle missing values; developing fuzzy models using clustering-based rule generation; implementing an adaptive neuro-fuzzy inference systems; and exploring deep learning architectures, such as Long Short-Term Memory (LSTM) networks, to capture temporal dependencies.

Finally, the performance of all models, fuzzy, neuro-fuzzy, and deep learning, is compared against baseline regression approaches using established error metrics. The objective is to assess how adaptive and interpretable AI-driven techniques can improve the accuracy and reliability of air quality forecasting under real-world data conditions, thereby enhancing the reliability of environmental monitoring and supporting proactive public health management.

2 Data Evaluation and Processing

The Air Quality dataset from the UCI Machine Learning Repository contains hourly measurements of various atmospheric pollutants and meteorological parameters collected between March 2004 and February 2005. This dataset is widely used for pollution modeling tasks due to its continuous temporal coverage and multiple correlated pollutant and sensor variables. The original dataset includes 15 numerical attributes: `Date`, `Time`, `CO(GT)`, `PT08.S1(CO)`, `NMHC(GT)`, `C6H6(GT)`, `PT08.S2(NMHC)`, `NOx(GT)`, `PT08.S3(NOx)`, `NO2(GT)`, `PT08.S4(NO2)`, `PT08.S5(O3)`, `T`, `RH`, and `AH`.

2.1 Handling Missing and Invalid Values

Within the dataset, missing or invalid readings are encoded as -200. These values were first identified and replaced using interpolation techniques to maintain temporal continuity in the time series. Linear interpolation was selected because it provides a good trade-off between smoothness and computational simplicity for time-series data.

Table 1: Number of Missing Values per Variable

Variable	Missing Values
Date	0
Time	0
CO(GT)	1683
PT08.S1(CO)	366
NMHC(GT)	8443
C6H6(GT)	366
PT08.S2(NMHC)	366
NOx(GT)	1639
PT08.S3(NOx)	366
NO2(GT)	1642
PT08.S4(NO2)	366
PT08.S5(O3)	366
T	366
RH	366
AH	366

2.2 Daily and Monthly Evaluation

To assess data stability, daily and monthly averages were computed for all numerical features as seen in figure 1. This evaluation helped visualize long-term trends and detect potential anomalies or sensor drifts over the recording period. Based on these trends alone, no clear evidence of seasonality could be inferred across the variables, suggesting that external periodic influences were limited or obscured by sensor variability.

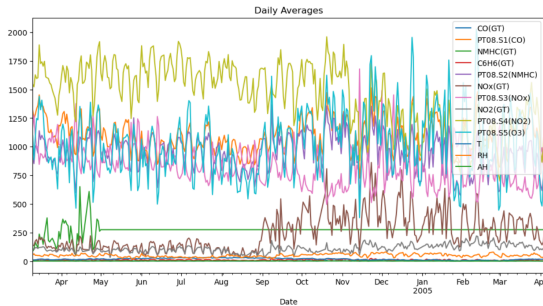


Figure 1: Daily average trends of selected pollutants before cleaning.

2.3 Flatline Detection and Data Filtering

Upon inspecting the time-series plots, extended periods of flatline behaviour were identified, as illustrated

in Figure 2, particularly in variables where more than 15% of the data had been interpolated. This indicated that interpolation over large gaps could introduce artificial patterns not present in the real measurements. To preserve the dataset's integrity, all segments exhibiting excessive interpolation were removed.

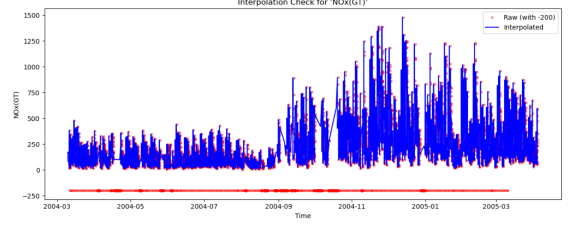


Figure 2: Example of flatline behaviour detected after heavy interpolation.

2.4 Final Dataset Selection

After filtering, only the most reliable and informative attributes were retained. The final selected features were:

- Date, Time
- PT08.S1(CO)
- C6H6(GT)
- PT08.S2(NMHC)
- PT08.S3(NOx)
- PT08.S4(NO2)
- PT08.S5(O3)
- T, RH, AH

These attributes were retained based on their data quality, stability, and predictive relevance. The resulting dataset provides a clean, consistent foundation for subsequent fuzzy, neuro-fuzzy, and deep learning models.

2.5 Data Preparation for Modeling

After the cleaning and feature selection process described above, the resulting dataset was prepared for modeling. The preprocessing applied to the data consisted of standardization and dataset splitting for training, testing, and validation.

Standardization was performed to center each feature around zero and scale it to have unit variance. This ensures that all variables contribute equally during model training and prevents features with larger magnitudes from dominating the learning process.

The standardized data were then divided into training and testing subsets. For neural-based approaches, an additional validation set was also created to monitor generalization performance during training and to help prevent overfitting.

Since neural networks, particularly recurrent architectures, require sequential input, the data were further transformed into overlapping temporal sequences. Each sequence represents a sliding window of past observations used to predict future values. This approach allows the model to capture temporal dependencies and patterns inherent in the air quality data.

3 Features

After preprocessing, the dataset was reduced to a subset of the most consistent and informative variables. The retained attributes include sensor readings and meteorological parameters that provide contextual information for pollutant concentration estimation. The **Date** and **Time** variables were used solely for temporal indexing and data alignment, and were not included as model inputs. The final predictive features are described below:

- PT08.S1(CO) — Sensor channel primarily sensitive to reducing gases such as CO.
- PT08.S2(NMHC) — Sensor channel responsive to non-methane hydrocarbons.
- PT08.S3(NO_x) — Sensor channel primarily sensitive to nitrogen oxides (NO and NO₂).
- PT08.S4(NO₂) — Sensor channel mainly responsive to oxidizing gases, particularly NO₂.
- PT08.S5(O₃) — Sensor channel sensitive to ozone (O₃) and other oxidants.
- T — Ambient temperature (°C).
- RH — Relative humidity (%).
- AH — Absolute humidity.

Following the approach proposed in the original study by De Vito [3], the reference benzene concentration (C6H6(GT)) was selected as the prediction target. Benzene is a hazardous air pollutant with well-documented carcinogenic effects, even at low concentrations, making it a critical indicator of air quality and public health risk.

The selected sensor and meteorological variables collectively capture the physical and chemical conditions influencing benzene concentration in the atmosphere. These inputs allow the models to learn complex nonlinear relationships and temporal dependencies between environmental factors and pollutant behaviour.

4 Evaluation of Each Model

4.1 Fuzzy Model Optimized with Least Squares

The Takagi–Sugeno–Kang (TSK) fuzzy model was implemented as a baseline due to its interpretability and robustness when handling noisy or uncertain data.

Fuzzy rules were generated automatically using Fuzzy C-Means clustering, which produced five clusters. The corresponding cluster centers and spreads were used to initialize Gaussian membership functions for each input. The fuzzy partition coefficient (FPC = 0.575) indicated a reasonable degree of separation among the clusters.

The model structure includes a fuzzy inference layer that calculates how strongly each rule is activated and a linear output layer that combines these activations to produce the final prediction. After defining the membership functions, the output parameters of each rule were estimated using Least Squares regression, based on the relationship between the rule activations and the target output.

The data were standardized, and 80% of the samples were used for training. The model achieved a Root Mean Squared Error (RMSE) of 0.2196 on the test set, showing good predictive capability. As shown in figure 3 shows it captured the general variation in benzene concentration but tended to underestimate higher peaks and some outliers at lower concentrations. Despite its stability and interpretability, this approach lacks adaptability, as the membership functions remain fixed after initialization, limiting its ability to model more complex non-linear behaviours.

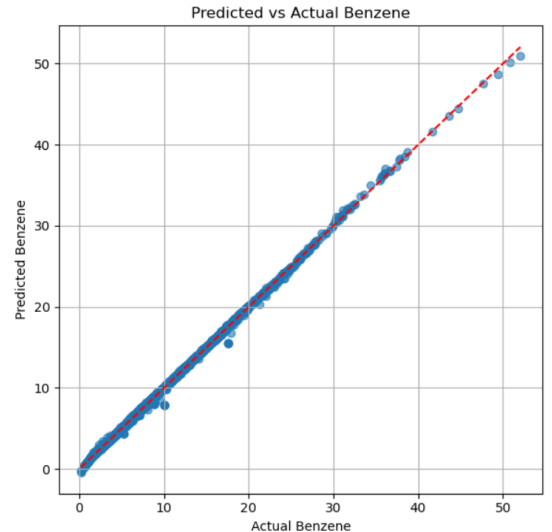


Figure 3: Least Square optimized model Scatter Plot.

4.2 ANFIS Model Optimization

4.2.1 Model Architecture

Both models are based on a first-order Takagi–Sugeno–Kang (TSK) Adaptive Neuro-Fuzzy Inference System (ANFIS) architecture. This structure was selected for its balance between interpretability and adaptability. Each input variable is represented by five Gaussian membership functions (MFs), which are combined into fuzzy rules. Each rule employs a first-order linear consequent, and the final system output is computed as the normalized weighted average of all rule outputs.

Conceptually, the ANFIS consists of five layers: fuzzification, rule (antecedent), normalization, consequent, and output aggregation. This layered design enables the model to capture nonlinear relationships while maintaining a transparent, rule-based representation.

It is worth noting that the ANFIS gradient descent (GD) optimization was implemented using the public GitHub library <https://github.com/mZaiam/ANFISpy.git>, based on the methodology described in [2].

4.2.2 Hybrid ANFIS Training

In the hybrid training strategy, the antecedent parameters (membership functions) were initialized using Fuzzy C-Means clustering. This clustering provides meaningful starting points for the MF centers and spreads, ensuring that the fuzzy rules appropriately cover the input space. Training alternates between two steps: gradient descent is applied to the antecedent parameters, while the consequent parameters (linear rule coefficients) are optimized analytically using least squares.

This hybrid approach exploits the linearity of the consequent parameters to obtain an exact solution, while allowing the nonlinear membership functions to be learned iteratively. The training was performed with a learning rate of 0.001, batch size of 32, and 10 hybrid iterations, each including 20 GD epochs. This configuration enables efficient convergence and high predictive accuracy.

4.2.3 Neural ANFIS Training

In contrast, the neural ANFIS implementation treats the entire system as a fully trainable neural network, where all parameters, including membership functions and rule consequents, are optimized simultaneously using backpropagation and the Adam optimizer. No clustering-based initialization was used, and the membership functions were initialized with random values.

Although the architecture remains identical to that of the hybrid ANFIS, the purely neural training ap-

proach faces a more challenging optimization landscape, as the antecedent and consequent parameters are highly coupled and nonlinear in the loss function. Training was conducted for 100 epochs with a fixed learning rate of 0.001 and a batch size of 32, while monitoring both training and validation losses to ensure convergence as seen in figure 4.

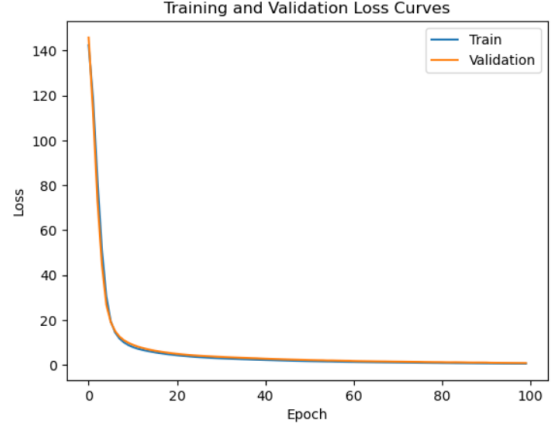


Figure 4: ANFIS GD optimization Loss Function.

4.2.4 Model Evaluation

Both the hybrid and neural ANFIS models were evaluated on different test datasets to assess predictive performance and generalization. This difference in test data may also contribute to some disparity in the results.

The hybrid ANFIS model, trained using a combination of gradient descent for the antecedent parameters and least squares for the consequents, achieved a test RMSE of 0.1911. As shown in Figure 5, the predicted benzene concentrations closely follow the measured values, with only minor deviations at higher concentrations and a few outliers at lower levels. The structured initialization using Fuzzy C-Means clustering allowed the model to converge efficiently and capture the nonlinear patterns in the data.

In contrast, the neural ANFIS model, trained end-to-end using gradient descent without clustering-based initialization, achieved a higher test RMSE of 0.7982. Figure 6 illustrates that while the model captured general trends, its predictions are less accurate and exhibit larger deviations from the true values. This behaviour is attributed to the more challenging optimization landscape when all parameters are trained simultaneously, without the benefit of alternating least-squares optimization for the consequents.

These results highlight that although both models share the same underlying architecture, the hybrid training strategy provides superior convergence speed

and predictive accuracy compared to purely gradient-based optimization.

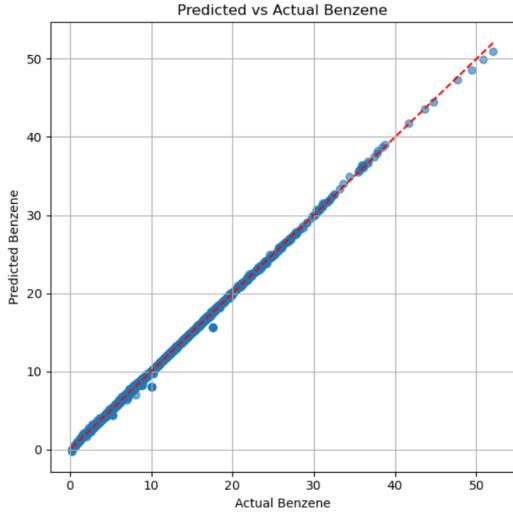


Figure 5: ANFIS hybrid optimization scatter plot.

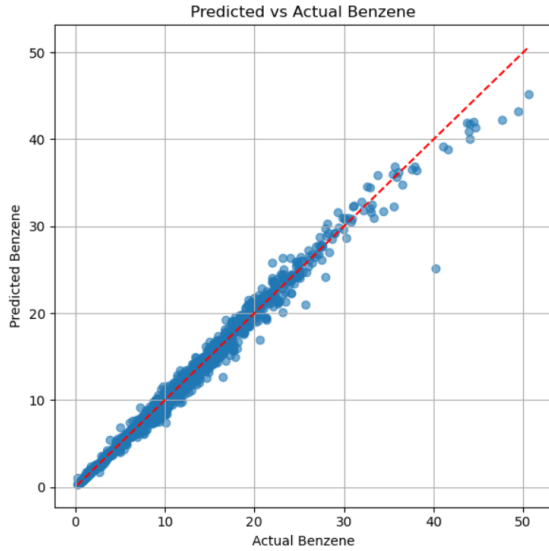


Figure 6: ANFIS GD optimization scatter plot.

4.3 LSTM Model

4.3.1 Model Architecture

The fourth model implemented was a two-layer Long Short-Term Memory (LSTM) network designed to capture the temporal dynamics present in the air quality data. In contrast to the fuzzy and neuro-fuzzy models, the LSTM focuses on learning sequential dependencies from past sensor readings to predict future benzene concentrations. The network was configured with a hidden size of 128 units per layer.

4.3.2 Model Training

Input sequences were constructed using sliding windows of 10 time steps, which served as inputs to the network. The model was trained for 20 epochs using the Adam optimizer and mean squared error (MSE) loss function, with a batch size of 32 and a fixed learning rate of 0.001. This configuration was selected to balance model complexity, convergence speed, and generalization capability.

During training, the model exhibited consistent improvement in performance across epochs, as shown in Figure 7. Both training and validation losses decreased substantially before stabilizing near the final epochs, indicating that the model effectively learned temporal dependencies without overfitting.

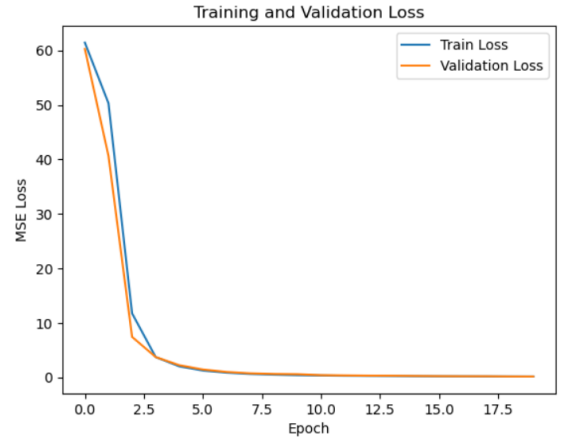


Figure 7: LSTM loss function during training.

4.3.3 Model Evaluation

After training, the LSTM model achieved a test RMSE of 0.3225. As illustrated in Figure 8, the predicted benzene concentrations closely follow the overall trajectory of the measured values, particularly at lower concentration levels. However, at higher concentrations, the model tends to deviate significantly, which contributes to the observed RMSE value.

With further hyperparameter optimization or the inclusion of longer temporal sequences, the LSTM's predictive performance could likely be further improved, potentially approaching that of the hybrid ANFIS model.

4.4 ANFIS-CNN-LSTM Model

4.4.1 FL-CNN-LSTM Model Architecture

The FL-CNN-LSTM architecture integrates fuzzy logic with convolutional and recurrent neural network components to effectively model the nonlinear and time-dependent behaviour of atmospheric pollutant

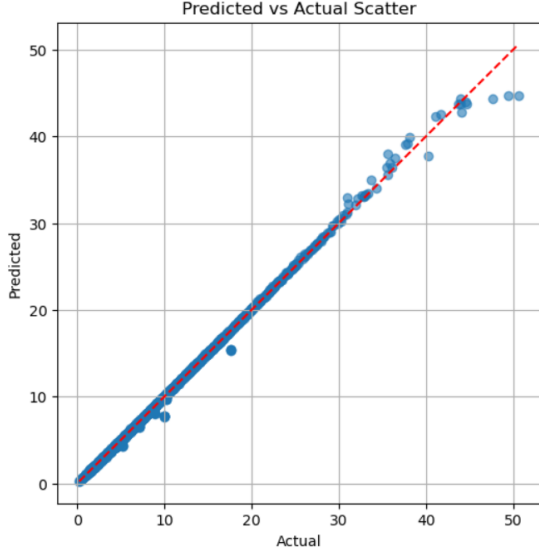


Figure 8: LSTM model scatter plot comparing predicted and measured benzene concentrations.

data. The structure was adapted from [1], serving as a strong baseline given the similarity between both problems, where the original work focuses on indoor air quality while the present project addresses outdoor conditions.

The model begins with a fuzzy layer that applies Gaussian membership functions to each input variable, converting continuous sensor and meteorological measurements into fuzzy linguistic representations. These memberships act as additional descriptive features, allowing the network to capture uncertainty and gradual transitions in pollutant levels. This feature enrichment enhances the downstream learning process, particularly when dealing with noisy or overlapping environmental conditions.

The subsequent one-dimensional convolutional (1D-CNN) layers extract local temporal features and inter-variable relationships from the fuzzy-enhanced input sequences. These layers are particularly effective for identifying short-term variation patterns and spatial correlations among sensor readings.

Finally, the extracted temporal features are passed to a Long Short-Term Memory (LSTM) layer, which captures long-term dependencies and trends in the data. The LSTM's gated structure enables the model to retain relevant temporal information while mitigating the effects of vanishing gradients, thereby improving prediction stability over extended time horizons.

Overall, this architecture combines the interpretability of fuzzy logic, the feature extraction capability of CNNs, and the sequence-learning strength of LSTMs. This makes it well suited for air quality forecasting tasks, where complex temporal interactions and measurement uncertainty play a significant role.

4.4.2 Model Training

The model was trained with a fixed learning rate of 0.001, a batch size of 32, a sequence length of 10, and for a total of 200 epochs. The fuzzy layer was initialized with five Gaussian membership functions.

As shown in Figure 9, the loss function for both training and validation data exhibits a rapid decline during the initial epochs, indicating effective learning and consistent performance improvement. Overall, the model demonstrates stable convergence, suggesting that the chosen architecture and training configuration were effective in minimizing prediction error.

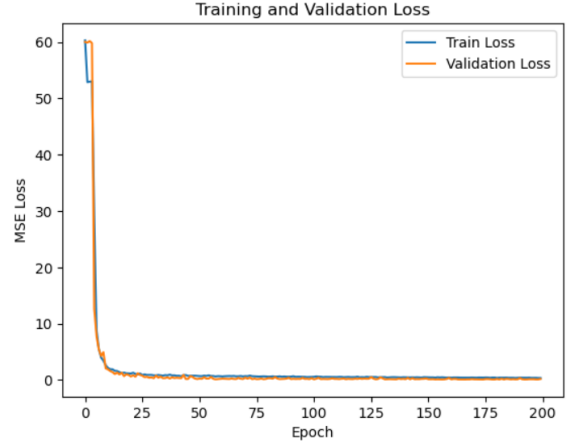


Figure 9: FL-CNN-LSTM model loss function.

4.4.3 Model Testing

The model achieved a Root Mean Squared Error (RMSE) of 0.4149. With further hyperparameter optimization and extended training, this error could likely be reduced further, as the training RMSE continued to decrease during the final epochs.

As shown in Figure 10, the model tends to underestimate higher benzene concentrations, suggesting that these deviations may be influenced by noise or unreliable sensor readings. Incorporating an outlier detection or rejection step during the preprocessing stage could therefore enhance model performance.

5 Results Assessment

Overall, the findings show that the hybrid models, and especially the ANFIS approach, achieved the most balanced trade-off between accuracy and interpretability. Deep learning architectures, such as LSTM and FL-CNN-LSTM, were able to capture the temporal behaviour of the data, but their improvement in predictive performance was modest. This is likely due to the limited amount of data and the short temporal span used during training. In these circumstances, simpler

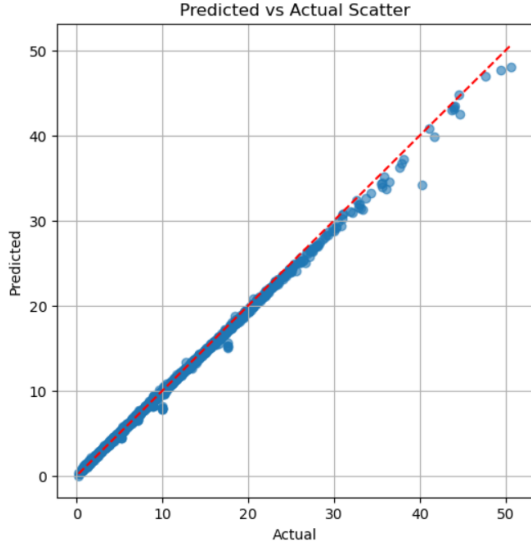


Figure 10: Predicted vs Actual scatter plot for the FL-CNN-LSTM model.

fuzzy systems often generalize better, as they are less sensitive to noise and data fluctuations. These results underline that adding complexity to a model does not necessarily lead to better outcomes, particularly when data availability and variability are restricted.

Table 2: Model performance comparison based on RMSE.

MODEL	RMSE
Fuzzy TSK (Least Squares)	0.2196
ANFIS	0.1911
ANFISpy	0.7982
LSTM	0.3225
FL-CNN-LSTM	0.4149

6 Conclusion

In conclusion, the comparative analysis of all implemented models shows that the hybrid approaches, particularly the ANFIS architecture, achieved the best balance between predictive accuracy, interpretability, and computational stability. By combining the adaptive learning capability of neural networks with the rule-based transparency of fuzzy logic, ANFIS effectively captured nonlinear relationships within the dataset while maintaining a structure that remains understandable and traceable. This makes it particularly suitable for environmental applications, where the interpretability of the model is as important as its predictive performance.

Deep learning models such as LSTM and FL-CNN-LSTM were able to identify temporal dependencies and complex relationships among variables, but their improvements in predictive accuracy were relatively

modest. This outcome can be explained by several factors. First, neural networks typically require large, diverse, and low-noise datasets to reach their full potential, whereas the dataset used in this project was limited and presented some variability in sensor measurements. Second, the temporal window used during training was short, restricting the models' ability to capture long-term dependencies. Third, deep architectures tend to overfit when data are scarce, as they may memorize local fluctuations rather than learning stable and generalizable patterns. As a result, although these models can represent complex structures, their performance may degrade when the data are limited or inconsistent.

In contrast, simpler fuzzy models demonstrated strong robustness to noise and missing data, thanks to their design based on linguistic rules that allow them to handle uncertainty in a more stable and interpretable way. Their performance remained consistent despite the dataset's limitations, showing that interpretability and reliability can be more valuable than computational complexity in data-constrained scenarios. These results emphasize that increasing model complexity does not guarantee better performance; on the contrary, it can heighten sensitivity to data imperfections and instability during training.

It is also important to note that most of the training experiments in this study were conducted with fixed hyperparameter settings, without extensive tuning or cross-validation. Deep learning architectures, in particular, are highly sensitive to hyperparameters such as learning rate, sequence length, and network depth. Therefore, it is plausible that additional optimization could further improve their predictive performance. Nonetheless, the current results provide a clear indication of each model's relative strengths and limitations under comparable training conditions.

Overall, this project suggests that hybrid intelligent systems, especially those integrating fuzzy reasoning with adaptive learning, represent a promising middle ground. They combine the interpretability and robustness of fuzzy logic with the flexibility of neural learning, offering a practical and transparent framework for air quality forecasting and other real-world environmental prediction tasks.

For inspection of the code, it is available at <https://github.com/Afonso-Duarte/Intelligent-Systems-Project.git>.

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